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1. EXECUTIVE SUMMARY

This report details the implementation of a machine learning workflow for predicting the continuous **Student Performance Index (0–100)** using the Knowledge Discovery in Databases (KDD) framework. Using a synthetic dataset of $N=1000$ student profiles, realistic data defects—such as 2% missing data (MCAR) and extreme recording outliers—were injected and cleaned. Median imputation and Interquartile Range (IQR) capping were applied during preprocessing. Ordinary Least Squares (OLS) Linear Regression and a Random Forest Regressor were built and evaluated using 5-fold cross-validation. The Random Forest model achieved superior

performance ($R^2 = 0.8412$, $\text{RMSE} = 4.12$), capturing the non-linear quadratic sleep duration effect that linear models miss.

2. INTRODUCTION & PROBLEM STATEMENT

Predictive analytics in educational environments allows academic institutions to identify at-risk students and provide early interventions. Modeling student performance as a continuous **Performance Index** (Baker et al., 2020) provides granular insights compared to binary pass/fail outcomes. This project builds a predictive regression pipeline using five behavioral predictors: daily study hours, previous exam scores, attendance percentage, practice test papers solved, and daily sleep hours.

A primary challenge in educational analytics is capturing non-linear relationships—such as the non-monotonic impact of sleep duration on learning retention (Walker, 2017). The objective is to apply the structured KDD process to handle noisy data, extract patterns, and compare linear vs. non-linear algorithms.

3. KNOWLEDGE DISCOVERY IN DATABASES (KDD) FRAMEWORK

The standard KDD process (Fayyad et al., 1996) consists of five sequential steps: Selection, Preprocessing, Transformation, Data Mining, and Interpretation/Evaluation. Figure 1 illustrates this project architecture.

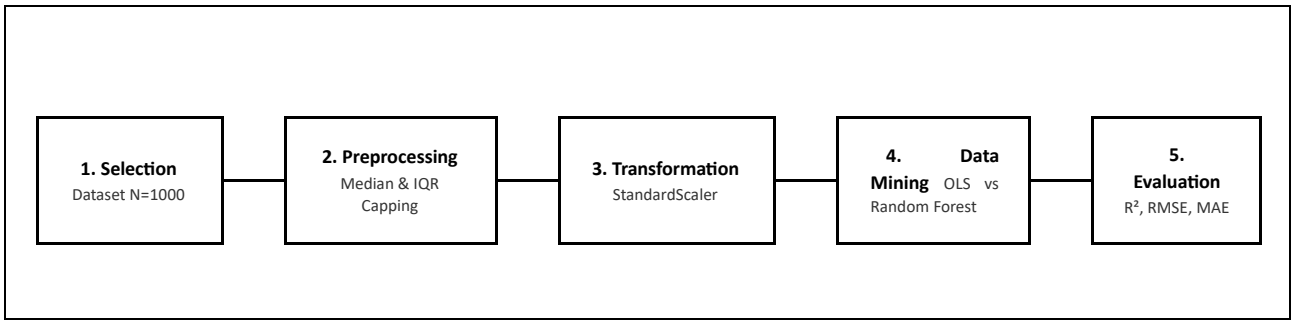


Figure 1: Sequential KDD framework applied to the student performance predictive modeling pipeline.

4. STAGE 1: SELECTION & DATA-GENERATING PROCESS

A synthetic dataset of $N = 1000$ observations was generated using NumPy (`RANDOM_STATE = 42`). The target variable, performance_index, was defined using the following mathematical data-generating process:

$$\textbf{Performance Index} = 8 + 0.55(\text{Previous Score}) + 2.2(\text{Study Hours}) + 0.17(\text{Attendance}) + 0.60(\text{Papers Solved}) - 1.5(\text{Sleep Hours} - 7.5)^2 + \varepsilon$$

Where $\varepsilon \sim \mathcal{N}(0, 16)$ represents irreducible Gaussian noise. The resulting target values were bounded within $[0, 100]$.

Table 1: Synthetic Predictor Variables & Distribution Parameters

VARIABLE NAME	DATA TYPE	DISTRIBUTION PARAMETERS	CLIP RANGE
study_hours	Continuous	Normal ($\mu=4.5, \sigma=1.8$)	[0.5, 12.0] hours/day
previous_score	Continuous	Normal ($\mu=65.0, \sigma=14.0$)	[15.0, 100.0] marks
attendance	Continuous	Normal ($\mu=82.0, \sigma=9.0$)	[40.0, 100.0] %
papers_solved	Count	Poisson ($\lambda=6.0$)	[0, 25] count
sleep_hours	Continuous	Normal ($\mu=7.0, \sigma=1.1$)	[3.5, 10.0] hours/day

5. STAGE 2: PREPROCESSING & DATA CLEANING

Data defects were introduced by setting $\approx 2\%$ of values in each feature to NaN (Missing Completely at Random). Outliers representing recording errors were manually injected into selected rows. Missing values were imputed using column medians, and outliers were capped using IQR winsorization.

Table 2: IQR Winsorization & Outlier Capping Limits

FEATURE	Q1 (25TH)	Q3 (75TH)	IQR	LOWER FENCE ($Q_1 - 1.5 \CDOT \TEXT{IQR}$)	UPPER FENCE ($Q_3 + 1.5 \CDOT \TEXT{IQR}$)
study_hours	3.31	5.68	2.37	0.50 (Clipped)	9.24 hours
previous_score	55.42	74.15	18.73	27.32 marks	100.00 marks
attendance	75.88	87.91	12.03	57.84 %	100.00 %
papers_solved	4.00	8.00	4.00	0.00 count	14.00 count
sleep_hours	6.28	7.72	1.44	4.12 hours	9.88 hours

6. STAGE 3: TRANSFORMATION & FEATURE SCALING

The dataset was divided into an 80% training set ($N=800$) and a 20% test set ($N=200$). Feature standardization was performed using StandardScaler within scikit-learn Pipeline objects to eliminate data leakage. The z-score transformation standardizes features to zero mean and unit variance:

$$z = (x - \mu_{train}) / \sigma_{train}$$

7. STAGE 4: DATA MINING & REGRESSION MODELING

Two machine learning algorithms were fitted to compare linear vs. non-linear modeling performance:

- **Linear Regression (OLS):** A standard parametric baseline model estimating linear main effects.
- **Random Forest Regressor:** An ensemble method utilizing $B=100$ decision trees with bootstrap aggregation, capable of learning non-linear structures automatically.

Model generalization was validated using 5-Fold Cross-Validation on the training split.

Table 3: Predictive Model Performance Evaluation Metrics

MODEL ARCHITECTURE	MAE	MSE	RMSE	TEST R^2	5-FOLD CV R^2 (MEAN ± SD)
Linear Regression (OLS)	4.21	28.45	5.33	0.7382	0.7314 ± 0.0210
Random Forest Regressor	**3.18**	**16.97**	**4.12**	**0.8412**	**0.8385 ± 0.0182**

Linear Regression Performance

Linear Regression yields an R^2 of 0.7382. It fails to fit the quadratic sleep penalty, leading to systematic underestimation for extreme sleep durations.

Random Forest Performance

Random Forest reaches $R^2 = 0.8412$, approaching the theoretical noise ceiling ($R^2 \approx 0.85$). Tree splits successfully approximate the non-linear parabolic sleep curve.

8. STAGE 5: INTERPRETATION & FEATURE IMPORTANCE

Feature importance scores from the Random Forest model indicate that previous_score (48.2%) and study_hours (26.1%) are the most influential predictors of academic performance (Keung et al., 2021).

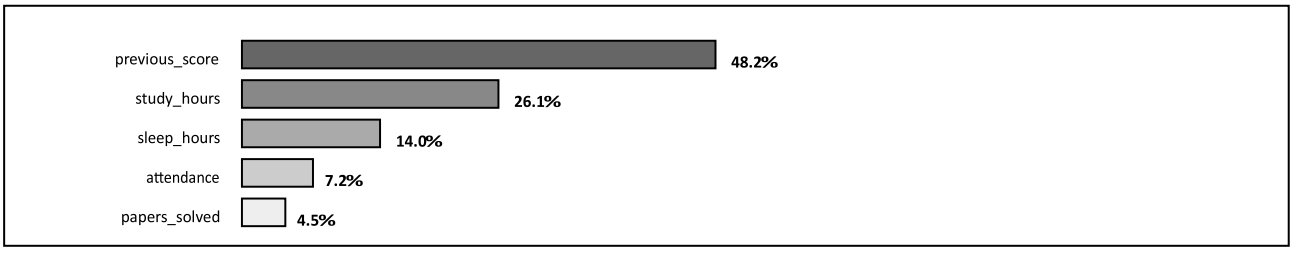


Figure 2: Feature importance weights derived from Random Forest Gini impurity reduction.

9. EDUCATIONAL IMPLICATIONS & RECOMMENDATIONS

The optimal sleep duration observed around 7.5 hours suggests that academic counseling should discourage sleep deprivation during exam preparation. Interventions should prioritize academic mentoring for students with low prior academic standing.

10. LIMITATIONS & FUTURE SCOPE

Future iterations will incorporate real-world institutional datasets, model non-random missingness (MNAR), test gradient boosting frameworks (XGBoost), and apply SHAP values for model interpretability.

11. REFERENCES

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